

M. Fiona Molloy

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EDUCATION & EMPLOYMENT

Postdoctoral Fellow, University of Michigan, Ann Arbor, MI

Addiction Center, Department of Psychiatry
Michigan Institute for Data and AI in Society (MIDAS) Postdoctoral Affiliate
Mentors: Dr. Chandra Sripada, Dr. Alex Weigard, and Dr. Mary Heitzeg
NIAAA T32 Postdoctoral Research Training Fellowship (T32 AA007477-29)

Ph.D. (Cognitive Neuroscience), The Ohio State University, Columbus, Ohio

Co-Advisors: Dr. David Osher and Dr. Zeynep Saygin
Dissertation: Individualized Brain Mapping in Infants, Adolescents, and Adults
December 2023

Master of Applied Statistics, The Ohio State University, Columbus, Ohio

Qualifying exam passed January 2020
Autumn 2020

Master of Science (Cognitive Psychology), The Ohio State University, Columbus, Ohio

Thesis: An Integrative Model of Response Inhibition
Autumn 2020

Bachelor of Science, The Ohio State University, Columbus, Ohio

Major: Behavioral/Systems Neuroscience
Minor: Substance Misuse and Addiction; Minor: Integrative Approaches to Health and Wellness
Honors: Research Distinction in Neuroscience
Spring 2018

Awards

MIDAS Affiliate Support Funding | University of Michigan
Summer Research Excellence Award | The Ohio State University
H. Dean and Susan Regis Gibson Research Award | The Ohio State University | \$3,000
Elsevier/Vision Research Virtual Travel Award | Vision Sciences Society
SBS Graduate Scholarship Award | The Ohio State University
University Fellowship | The Ohio State University | \$25,800
Prominence Scholarship | The Ohio State University | \$111,162 (4 year tuition)
Summer Undergraduate Research Fellowship | The Ohio State University | \$3,500
Second-Year Transformational Experience Program Fellowship | The Ohio State University | \$2,000
Women's Board Scholarship | MedStar Montgomery Hospital | \$3,500

GRANTS

1F31HD107961-01: Molloy (PI). Predoctoral Individual National Research Service Award (2022-2023).
"Brain basis and neural circuitry of habit-like attention in infants and adults." \$33,167.00 (Year 1)

PUBLICATIONS

- Molloy, M.F.**, Lee, T., Jonides, J., Zhang, H., Sellers, J., Heathcote, A., Sripada, C., and Weigard, A.S. (in press). Joint cognitive models reveal sources of robust individual differences in conflict processing. *Computational Brain & Behavior*.
- Weigard, A.S., **Molloy, M.F.**, Sripada, C., and Heathcote, A. (in press). The diffusion model's drift rate parameter primarily reflects efficiency, rather than speed, of evidence accumulation. *Psychonomic Bulletin & Review*.
- Molloy, M.F.**, Taxali, A., Angstadt, M., Toda-Thorne, K., McCurry, K. L., Weigard, A., Kardan, O., Lehrmann, C., Vens, J., Michael, C., Heitzeg, M. M., and Sripada, C. (2026). Decreased sleep linked longitudinally and directionally to alterations in the brain's intrinsic functional architecture. *Developmental Cognitive Neuroscience*, 78, 101668.
- Molloy, M.F.**, Taxali, A., Angstadt, M., Greathouse, T., Toda-Thorne, K., McCurry, K. L., Weigard, A., Kardan, O., Burchell, L., Dziubinski, M., Choi, J., Vandersluis, M., Michael, C., Heitzeg, M. M., and Sripada, C. (2025). Regional, but not brain-wide, graph theoretic metrics are robustly and reproducibly linked to general cognitive ability. *Cerebral Cortex*, 35(4), bhaf074.
- Michael, C., Taxali, A., Angstadt, M., McCurry, K. L., Weigard, A., Kardan, O., **Molloy, M. F.**, Thorne, K., Burchell, L., Dziubinski, M., Choi, J., Vandersluis, M., Hyde, L. W., Heitzeg, M. M., and Sripada, C. (2025). Somatomotor disconnection links sleep duration with socioeconomic context, screen time, cognition, and psychopathology. *Biological Psychiatry: Global Open Science*, 100522.
- Kardan, O., Jones, N., Wheelock, M. D., Michael, C., Angstadt, M., **Molloy, M. F.**, Cope, L. M., Martz, M. M., McCurry, K. L., Hardee, J. E., Rosenberg, M. D., Weigard, A. S., Hyde, L. W., Sripada, C., and Heitzeg, M. M. (2025). Assessing neurocognitive maturation in early adolescence based on baby and adult functional brain landscapes. *Developmental Cognitive Neuroscience*, 73, 101543.
- Bibb, S.A., Yu, E.J., **Molloy, M.F.**, LaRocco, J., Resnick, P., Reeves, K., Phan, K. L., Krishna, S., and Saygin, Z.M. (2025). Pilot study comparing effects of infrared neuromodulation and transcranial magnetic stimulation using magnetic resonance imaging. *Frontiers in Human Neuroscience*, 19.
- Michael, C., Taxali, A., Angstadt, M., Kardan, O., Weigard, A., **Molloy, M. F.**, McCurry, K. L., Hyde, L. W., Heitzeg, M. M., and Sripada, C. (2024). Socioeconomic resources in youth are linked to divergent patterns of network integration and segregation across the brain's transmodal axis. *PNAS Nexus*, 3(9), 412.
- Molloy, M.F.**, Saygin, Z.M., and Osher, D.E. (2024). Predicting high-level visual areas in the absence of task fMRI. *Scientific Reports*, 14(1), 11376.
- Molloy, M. F.** and Osher, D. E. (2023). A personalized cortical atlas for functional regions of interest. *Journal of Neurophysiology*, 130(5), 1067–1080.
- Molloy, M. F.**, Yu, E. J., Mattson, W. I., Hoskinson, K. R., Taylor, H. G., Osher, D. E., Nelson, E. E., and Saygin, Z. M. (2023). Effect of extremely preterm birth on adolescent brain network organization. *Brain Connectivity*, 13(7), 394–409.
- Molloy, M.F.** and Saygin, Z.M. (2022). Individual variability in functional organization of the neonatal brain. *Neuroimage*, 253, 119101.
- Molloy, M.F.**, Romeu, R. J., Kvam, P.D., Finn, P.R., Busemeyer J.R., and Turner, B.M. (2020). Hierarchies improve individual assessment of temporal discounting behavior. *Decision*, 7, 212-224.
- Molloy, M.F.**, Bahg, G., Lu, Z.-L., and Turner, B. M. (2019). Individual differences in the neural dynamics of response inhibition. *Journal of Cognitive Neuroscience*, 31(12), 1976-1996.
- Molloy, M.F.**, Galdo, M., Bahg, G., Liu, Q., and Turner, B.M. (2019). What's in a response time?: On the importance of response time measures in constraining models of context effects. *Decision*, 6, 171-200.

Molloy, M.F., Bahg, G., Li, X., Steyvers, M., Lu, Z., and Turner, B.M. (2018). Hierarchical Bayesian analyses for modeling BOLD time series data. *Computational Brain & Behavior*, 2, 184-213.

Turner, B.M., Rodriguez, C.A., Liu, Q., **Molloy, M.F.**, Hoogendijk, M., and McClure, S.M. (2018). On the neural and mechanistic bases of self control. *Cerebral Cortex*, 29(2), 732-750.

MANUSCRIPTS SUBMITTED AND UNDER REVIEW

Molloy, M.F., Tomlinson, D.C., Cope, L.M., Sripada, C., Heitzeg, M.M., and Weigard, A.S. (*Under review*). Systematic decision-making during delay discounting, but not discounting rate, is robustly linked to psychopathology.

Lin, Z., **Molloy, M.F.**, Sripada, C., Kang, J., and Si, Y. (*Under review*). Population-weighted image-on-scalar regression analyses of large scale neuroimaging data.

Weigard, A.S., Hicks, B., **Molloy, M.F.**, Paige, K., McCurry, K., Cope, L., Hardee, J., Martz, M., Sripada, C., Zucker, R., Dinov, I., and Heitzeg, M. (*Under review*). Multivariate models predicting substance use disorder in young adults reveal two key developmental risk dimensions.

Schettini, E., **Molloy, M.F.**, and Saygin, Z.M. (*Submitted*). Bayesian modeling to extract single trials from blockwise fMRI data in developmental neuroimaging.

RESEARCH INTERESTS

- Computational models of brain and behavior, with a focus on neuroimaging and decision-making
- Dynamics of decision-making and other cognitive processes in addiction
- Neurocognitive predictors of substance use in adolescence
- Development of the connectome and higher-order brain networks from infancy to adulthood
- Application and development of robust statistical analyses and models for neuroscience and psychology that can account for individual differences

PRESENTATIONS

Talk Presentations

The University of Michigan. *Department of Psychology Methods Hour: Estimating joint distributions of cognitive model parameters to understand individual differences in conflict tasks.* Ann Arbor, MI. Spring 2024.

Center for Cognitive and Behavioral Brain Imaging (CCBBI) Research Day. *Identifying visual brain regions in the absence of task fMRI.* Columbus, OH, Autumn 2022 ****Awarded Best Presentation****

The University of Iowa. *Broadscale network organization from infancy to adulthood and predicting functional regions of interest.* Virtual, Summer 2022

Vision Sciences Society. *Identifying visual brain regions in the absence of task fMRI.* St. Pete Beach, FL, Spring 2022

Center for Cognitive and Behavioral Brain Imaging (CCBBI) Research Day. *A personalized cortical atlas, generated from individual subject voxelwise connectivity.* Columbus, OH, Autumn 2021

Chronic Brain Injury (CBI) Research Day. *Data Blitz: Innate organization of the human brain.* Columbus, OH, Spring 2021

(Cancelled due to COVID-19) **American Psychological Society Annual Convention,** *Hierarchies improve individual assessment of temporal discounting behavior.* Part of accepted symposium on “Unrelated or Unrelated? Sources of—and Solutions to—the mismeasurement pervasive across psychological science” organized by Nathaniel Haines, Spring 2020

Midwest Cognitive Science, *Hierarchical Bayesian models for understanding neural dynamics of response inhibition,* Columbus, OH, Spring 2019

Poster Presentations

- U-M Annual Data Science & AI Summit.** *Little Data Informs Big Data: The Longitudinal and Directional Impact of Reduced Sleep on Brain Networks.* Ann Arbor, MI, Autumn 2025
- Research Society on Alcohol Annual Meeting.** *Systematic Decision-making in Delay Discounting and Risk for Alcohol Use Disorder.* New Orleans, LA, Summer 2025
- Silverman Conference.** *Systematic Decision-Making in Delay Discounting is Closely Related to Cognition and Psychopathology.* Ann Arbor, MI, Spring 2025
- Cognitive Neuroscience Society Annual Meeting.** *Regional, but not brain-wide, graph theoretic metrics are robustly and reproducibly linked to general cognitive ability.* Boston, MA, Spring 2025
- Silverman Conference.** *Understanding individual differences in conflict tasks using joint cognitive modeling.* Ann Arbor, MI, Spring 2024
- U-M Neuroscience Conference.** *Contributions of the graphical properties of the functional connectome to general intelligence.* Ann Arbor, MI, Spring 2024
- Organization for Human Brain Mapping (OHBM) Annual Meeting.** *Predicting category-selective regions of interest using functional connectivity.* Montreal, Canada, Spring 2023
- Vision Sciences Society.** *A personalized cortical atlas for high level vision.* St. Pete Beach, FL, Spring 2022
- Society for Neuroscience Annual Meeting.** *A personalized cortical atlas, generated from individual subject voxelwise connectivity.* Virtual, Autumn 2021
- Vision Sciences Society.** *Innate organization of the human brain.* Virtual, Spring 2021
- Chronic Brain Injury (CBI) Research Day.** *Innate organization of the human brain.* Columbus, OH, Spring 2021
- Cognitive Neuroscience Society Annual Meeting,** *Best of both worlds: Integrating EEG and fMRI in the study of inhibition,* Virtual due to COVID-19, Spring 2020
- Society For Neuroscience Annual Meeting,** *Hierarchical Bayesian Analyses reveal individual differences in the neural dynamics of response inhibition,* Chicago, IL, Autumn 2019
- Cognitive Neuroscience Society Annual Meeting,** *Hierarchical Bayesian analyses for modeling BOLD time series data,* San Francisco, CA, Spring 2019
- Mathematical Psychology,** *Hierarchical Bayesian analyses for modeling BOLD time series data.* University of Wisconsin, Madison, WI, Summer 2018
- Midwest Cognitive Science,** *Hierarchical Bayesian analysis reveals complex neural dynamics of inhibitory control.* Indiana University, Bloomington, IN, Spring 2018
- The Richard J. and Martha D. Denman Undergraduate Research Forum,** *Hierarchical Bayesian analysis reveals complex neural dynamics of inhibitory control.* The Ohio State University Columbus, OH, Spring 2018
- Fall Undergraduate Research Forum,** *What's in a response time?* The Ohio State University, Columbus, OH, Autumn 2017
- The Richard J. and Martha D. Denman Undergraduate Research Forum,** *Neural basis of self control.* The Ohio State University Columbus, OH, Spring 2017

TEACHING

- Instructor of Record. *PSYCH 1100: Introduction to Psychology* (2021-2022)
- Graduate Course Assistant. *PSYCH 2220: Data Analysis in Psychology* (2020-2021)

SKILLS

- Resting state and task-based fMRI collection, preprocessing, and analysis (in FSL, FreeSurfer, SPM, MATLAB, and R: including modeling using GLM, hierarchical Bayesian models, multivariate dynamical systems, machine learning)
- Joint modeling of fMRI, EEG, behavior
- Cognitive modeling of choice and response time data
- EEG analysis (eeglab and FieldTrip), source localization
- Statistical analyses including: linear modeling, ANOVA, nonparametric methods, stochastic processes, both Bayesian and frequentist methods

- Programming in R, MATLAB, and Python
- Proficiency in LaTeX, Photoshop, JAGS, HTML

SERVICE

- Ad hoc reviewer: *Proceedings of the National Academy of Sciences*, *Neuroimage*, *Journal of Research on Adolescence*, *Behavior Research Methods*, *Scientific Reports*, *Neuropsychologia*
- Research mentor, Undergraduate Research Opportunity Program (UROP). Ann Arbor, MI (2025-present)
- Conference reviewer, Denman Undergraduate Research Forum. Columbus, Ohio (2022-2023)
- Center for Cognitive and Behavioral Brain Imaging (CCBBI) Diversity Committee. Columbus, Ohio (2020-2023)
- CCBBI Student Group. Columbus, Ohio (2018-2023)
- Conference planning committee for Midwest Cognitive Science. Columbus, Ohio (2019)

PROFESSIONAL DEVELOPMENT: RESEARCH

- Program in Biomedical Sciences (PIBS) 503: NIH-mandated Ethics and Responsible Conduct in Research (Autumn 2025)
- MICHR (Michigan Institute for Clinical and Health Research) – K Writing Workshop (Winter 2025)
- National Academies: Exploring the Bidirectional Relationship Between Artificial Intelligence and Neuroscience Workshop (Spring 2024)
- Computational Visual Neuroscience Laboratory Workshop and Retreat. Minneapolis, Minnesota (Winter 2022)
- Model-based Cognitive Neuroscience Summer School. Amsterdam, Netherlands (Summer 2018)

PROFESSIONAL DEVELOPMENT: TEACHING

- Center for Research on Learning and Teaching (CLRT) Course: “An Instructor's Guide to Promoting Student Mental Health and Well-Being.” University of Michigan (Autumn 2025, online course)
- CLRT Workshop: “Rubrics: Transparent and Efficient Assessment in Support of Students’ Learning.” University of Michigan (Summer 2025, online course)
- Teaching Psychology Seminar. The Ohio State University (Summer 2021, 5 credit hours)

MENTORSHIP

- Avni Rathi (UROP Undergraduate RA; Autumn 2025-present)
- Kyle Aspillera (UROP Undergraduate RA; Autumn 2025-present)
- Camille Lehrmann (Post-baccalaureate RA; Spring 2025-present)
- Josh Vens (Post-baccalaureate RA; Spring 2025-present)
- Lily Burchell (Post-baccalaureate RA; Autumn 2024-present)
- Maria Dziubinski (Post-baccalaureate RA; Autumn 2023-present)
- Katherine Toda-Thorne (Post-baccalaureate RA; Autumn 2023-Summer 2025)
- Jason Choi (Post-baccalaureate RA; Winter 2023-Summer 2025)
- Melanie Vandersluis (Post-baccalaureate RA; Autumn 2023-Summer 2025)
- Maggie Duffy (Undergraduate RA; Autumn 2022-Summer 2023)
- Emily Yu (Undergraduate RA; Summer 2021-Summer 2023)
- Noreen Altallaa (Undergraduate RA; Winter 2020)
- Stephanie Dietrich (Undergraduate RA; Winter 2020)